DisCo + the queen: closure gained, AUC paid

DNCn_QN mk2 on the endcap ABCD plane · R15 runs

Nathan Herling · UA ATLAS HEP · 30 Oct 2026

Two penalties, two jobs

DisCo

$$L = L_{\mathrm{BCE}} + \alpha\, \mathrm{dCor}^2(s_1, s_2 \mid \mathrm{bkg})$$

- Distance correlation over all background event pairs
- Pushes NN1 and NN2 toward full independence
- No cuts, no sign, no statistical floor

The queen (DNCn_QN mk2)

Each step: 9 random cut pairs in [0.3, 0.8]² (bkg quantile). At each cut k, closure error vs its statistical size:

$$z_k^2 = \frac{g_k^2}{F_k}, \qquad g_k = N_A - \frac{N_B\, N_C}{N_D}$$

QDR: penalizes each cut whose error exceeds statistics (per-cut noise floor).

$$L_{\mathrm{QDR}} = \frac{\beta}{K} \sum_{k=1}^{K} \max\left(0,\ z_k^2 - 1\right)$$

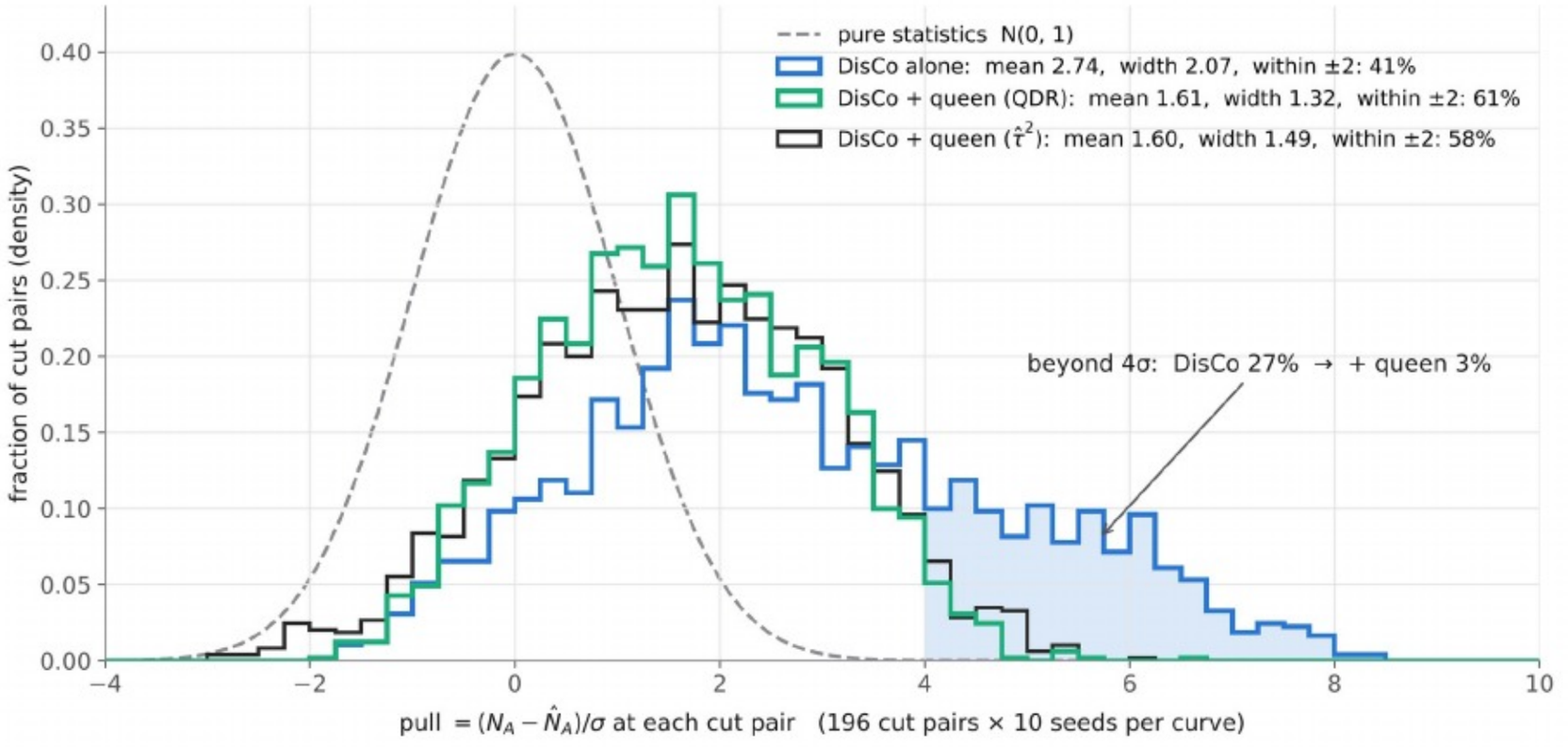
τ²: treats non-closure across cuts as a random variable; penalizes only its excess variance (pooled noise floor).

$$L_{\tau^2} = \beta\, \hat{\tau}^2, \qquad \hat{\tau}^2 = \frac{\max\left(0,\ \sum_k z_k^2 - K\right)}{\sum_k \sigma_{r,k}^{-2}}$$

$$\mathrm{pull} = \frac{N_A - \hat{N}_A}{\sqrt{N_A + \sigma^2_{\hat{N}_A}}}, \qquad \hat{N}_A = \frac{N_B\, N_C}{N_D}$$

Metric 1: the pull at each cut pair. With perfect closure, pulls follow N(0, 1): mean 0, width 1, 95% within ±2.

Metric 1: the queen pulls non-closure toward statistics



R15 confirm, DisCo α = 8 in every arm, seeds 3000–3009. What remains is mostly a +1.6σ offset, not cut-to-cut structure.

2.07 → 1.32

pull width (statistics: 1)

41% → 61%

cut pairs within ±2σ (statistics: 95%)

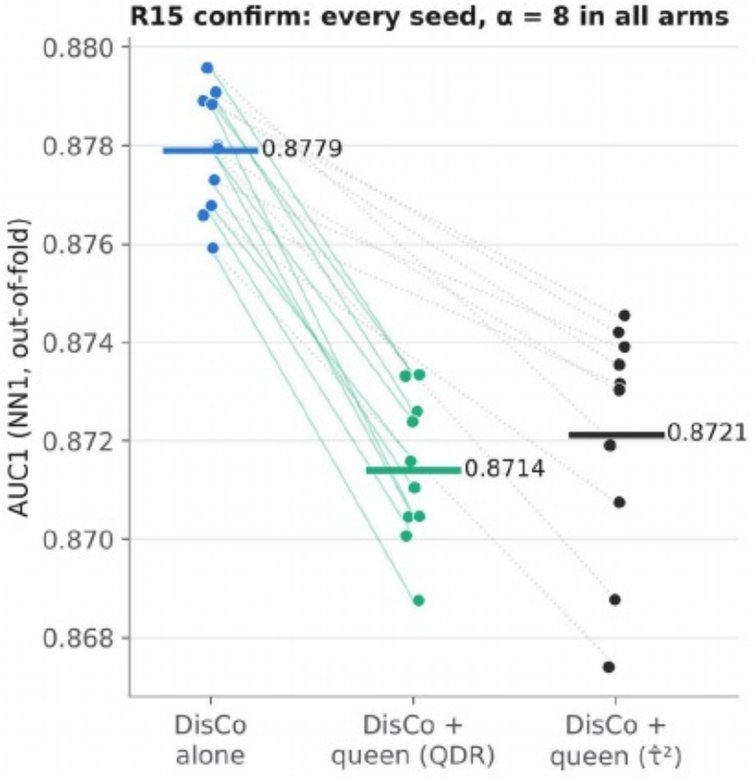
27% → 3%

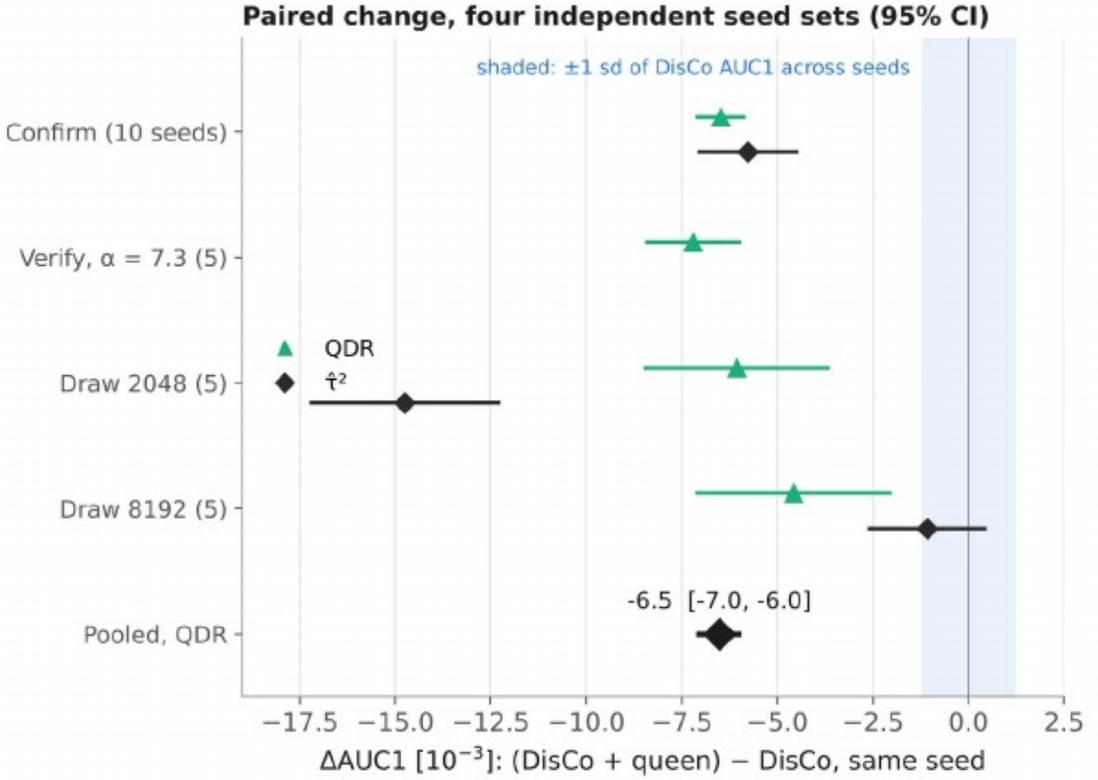
cut pairs beyond 4σ

Same seed, with vs without: width −0.72 [−0.88, −0.57], 10 of 10 seeds

Data: R15 confirm, endcap, seeds 3000–3009 · DisCo alone, QDR β 0.05, τ^2 · 3 arms × 10 seeds = 30 runs (30 ABCD planes)
Each run: 34,970 background + 9,996 signal events, out-of-fold · 196 cut pairs per plane → 5,880 pulls in all

Metric 2: the queen costs 0.0065 in AUC1





−0.0065
pooled ΔAUC1, 95% CI [−0.0070, −0.0060]

25 of 25
paired seeds lower, in 4 independent seed sets

5× seed sd
DisCo AUC1 varies by 0.0012 seed to seed

−0.011
signal efficiency at the WP (≈1 point)

Left: R15 confirm, seeds 3000–3009, 3 arms × 10 = 30 runs. Right: 25 paired seeds in 4 blocks: confirm 3000–3009 (10), verify 6000–6004 (5), draw 2048 and draw 8192 3100–3104 (5 + 5) → 70 runs · 34,970 bkg + 9,996 sig events per run

SLIDES 6–8 · A MYSTERY FROM THE SUMMER

DisCo's BCE curves don't look like the textbook ones

Since this summer, the signal and background score distributions of DisCo-trained networks have shown shapes that are not the textbook ones: a narrow spike in NN1, and background piled up against a score of 0 in NN2. The next three slides look at them, and propose a way to pull them apart.

6

What we see in score space

The shapes hold, except NN2 background. NN1 has one narrow window holding 7–9% of background, in every run.

7

The tool: read the curves in logit space

$z = \ln \frac{p}{1-p}$ spreads out what piles up at 0 and 1: scores of 0.007 and 3×10^{-7} sit 10 units apart.

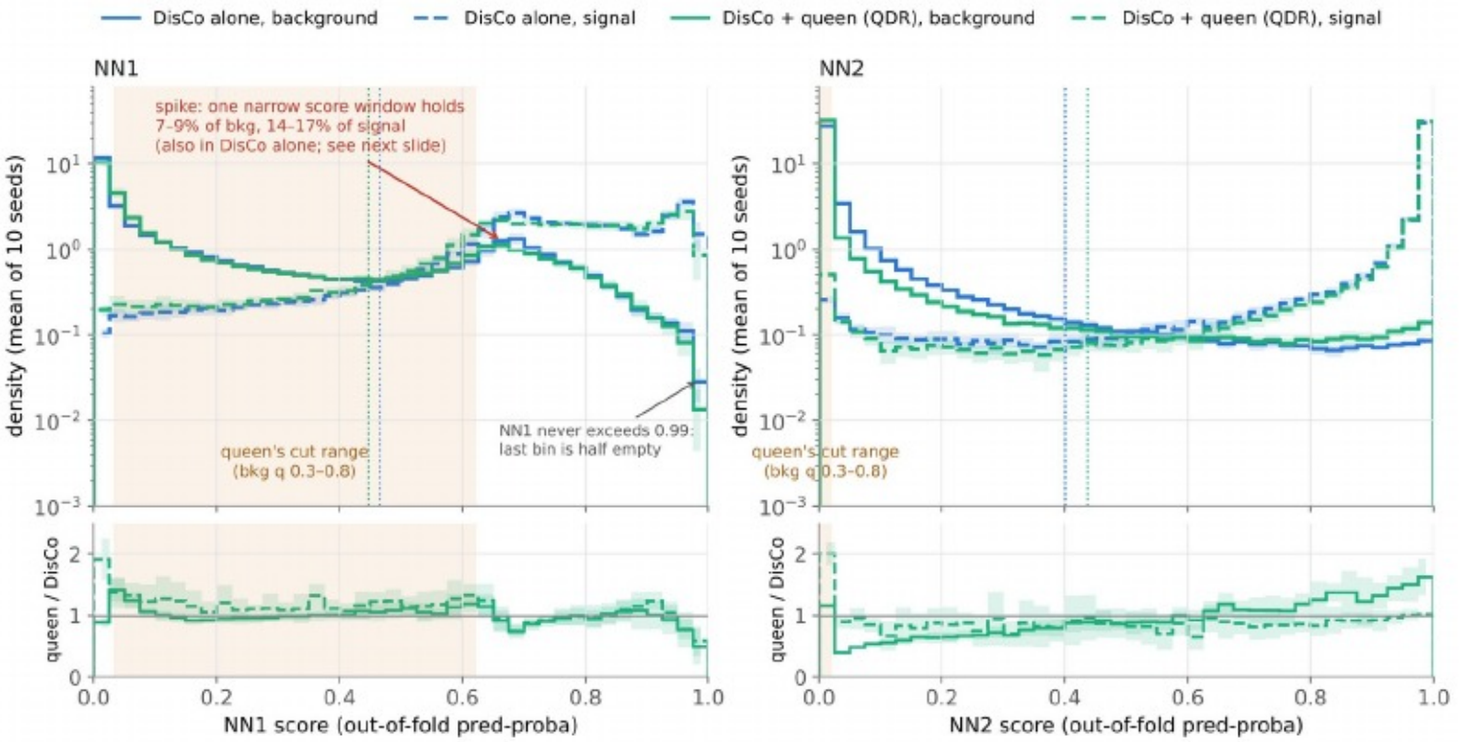
8

Pull one feature apart

NN1's spike: 74.5% of background vertices in it have no calo clusters, against 50.9% around it.

Data for slides 6–8: R15 confirm, endcap, seeds 3000–3009 · 34,970 background + 9,996 signal events per run, out-of-fold

Metric 3: shapes hold, except NN2 background



Band: ±1 sd across seeds. Dotted vertical: endcap WP cut (bkg quantile 0.70 for NN1, 0.946 for NN2), per arm. Ratio: per-seed queen/DisCo, mean ± sd. R15 confirm, α = 8, β = 0.05, seeds 3000–3009.

0.445 → 0.430

NN1 separation $\langle S^2 \rangle$, 10 of 10 seeds lower

0.860 → 0.855

NN2 separation $\langle S^2 \rangle$, 10 of 10 seeds lower

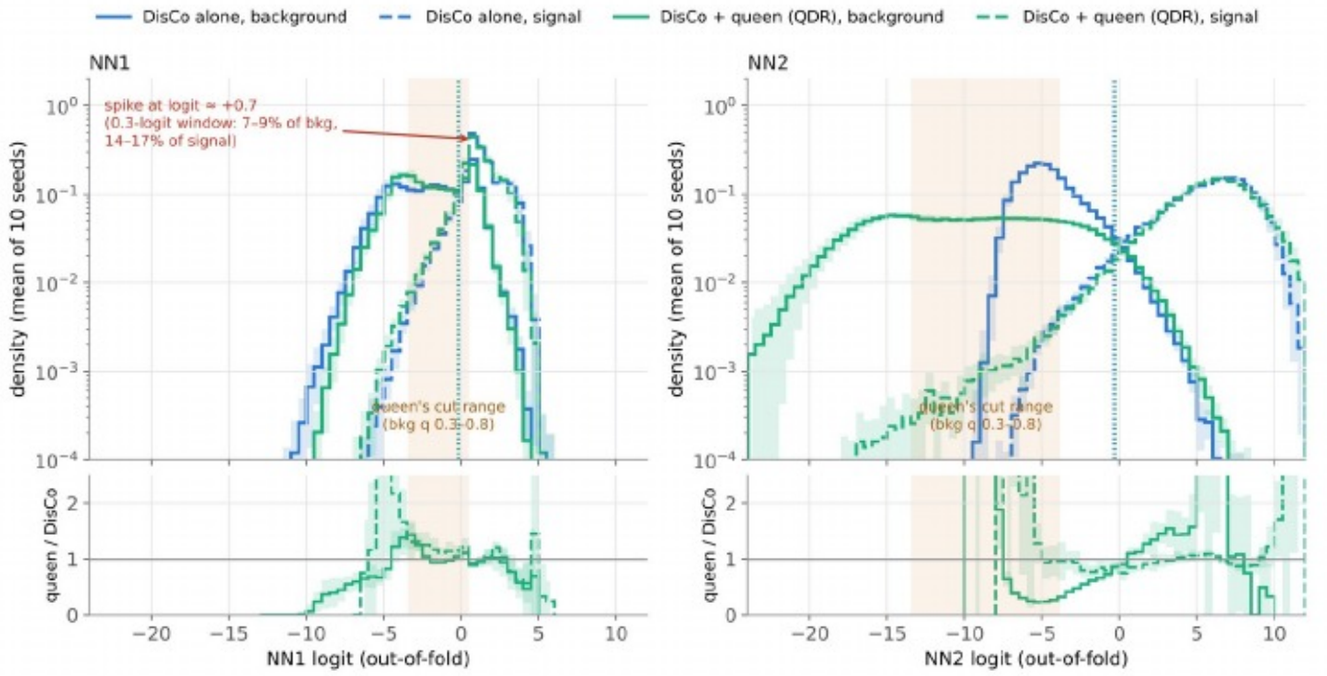
KS 0.59

NN2 background vs DisCo (seed-to-seed: 0.03)

Signal shapes match within seed scatter (KS ≤ 0.06). NN1 background shifts slightly (KS 0.07 vs 0.04).

Data: R15 confirm, endcap, seeds 3000–3009 · DisCo alone and QDR β 0.05 · 2 arms × 10 seeds = 20 runs
Each run: 34,970 background + 9,996 signal events, out-of-fold · curves are the 10-seed mean, band ±1 sd

In logit space, the queen reshapes NN2 background



Band: ±1 sd across seeds. Dotted vertical: endcap WP cut (bkg quantile 0.70 for NN1, 0.946 for NN2), per arm. Ratio: per-seed queen/DisCo, mean ± sd. R15 confirm, α = 8, β = 0.05, seeds 3000–3009.

$$z = \ln \frac{p}{1-p}, \qquad p = \sigma(z) = \frac{1}{1+e^{-z}}, \qquad m_k = \sigma(5\,(z - c_k))$$

Logit z: the network's raw output before the sigmoid; each unit multiplies the odds by e. The queen counts events through m_k, so her cuts live in z. Scores of 0.007 and 3e-7 look identical on slide 5 but sit 10 logit units apart here.

Possible NN2 pathology

2.9 → 9.6

NN2 bkg logit span over her cut range (q 0.3–0.8)

0.956 vs 0.967

NN2 bkg rank correlation: queen vs DisCo (same seed) against DisCo vs DisCo (next seed). Order barely changes: a stretch, not a reshuffle.

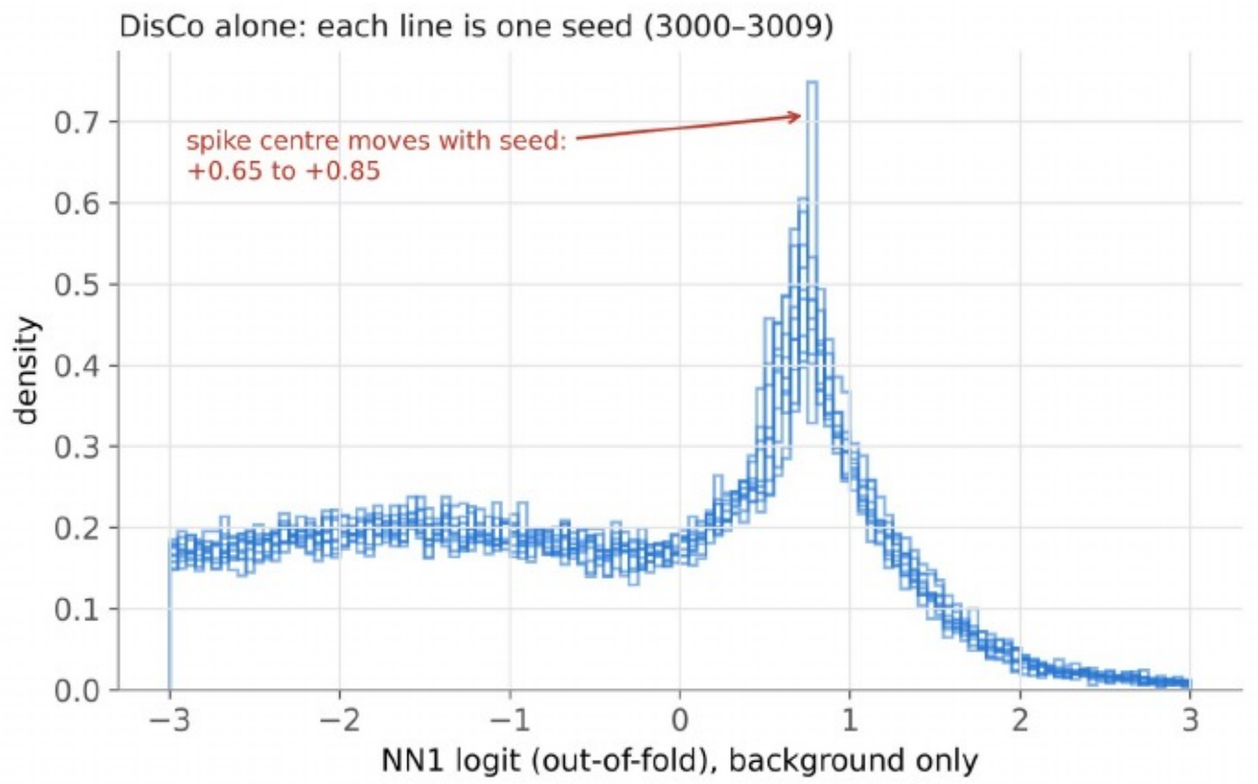
Why it could be pathological: her penalty only acts on events within ≈0.2 logit of a cut (the slope of m_k). The stretch leaves 3.3× fewer bkg events on each cut boundary (0.175 → 0.053 per logit unit), so the network can starve her gradient instead of fixing closure.

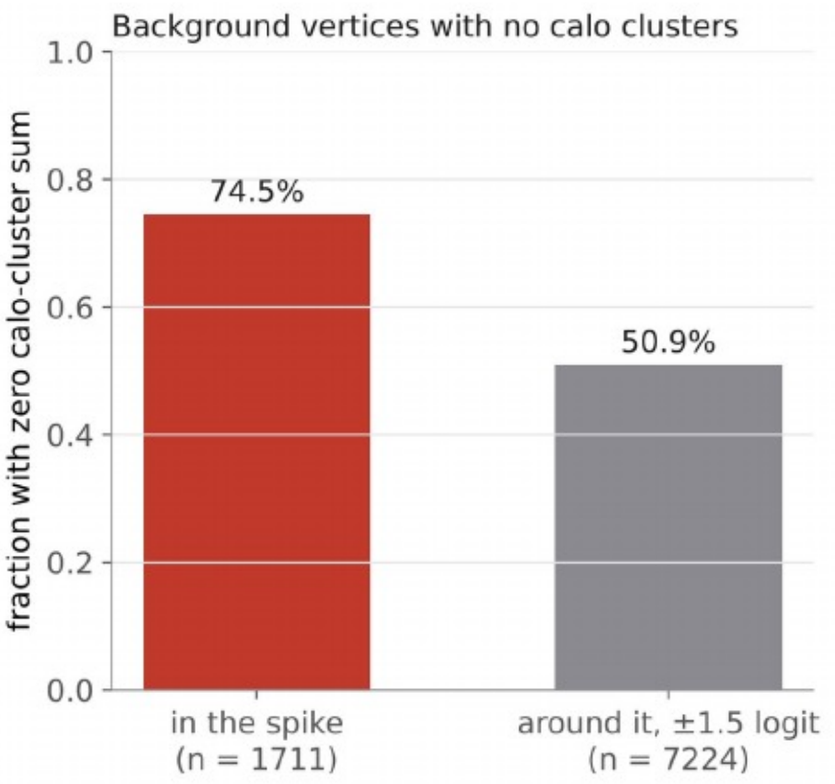
Against: hard-count closure improved; signal, AUC2 (0.9897 → 0.9891) and NN2 WP efficiency barely move.

Test: log her soft closure and the hard closure every epoch.

Data: R15 confirm, endcap, seeds 3000–3009 · DisCo alone and QDR β 0.05 · 2 arms × 10 seeds = 20 runs
Each run: 34,970 background + 9,996 signal events, out-of-fold · curves are the 10-seed mean, band ±1 sd

NN1's spike: vertices with no calo clusters





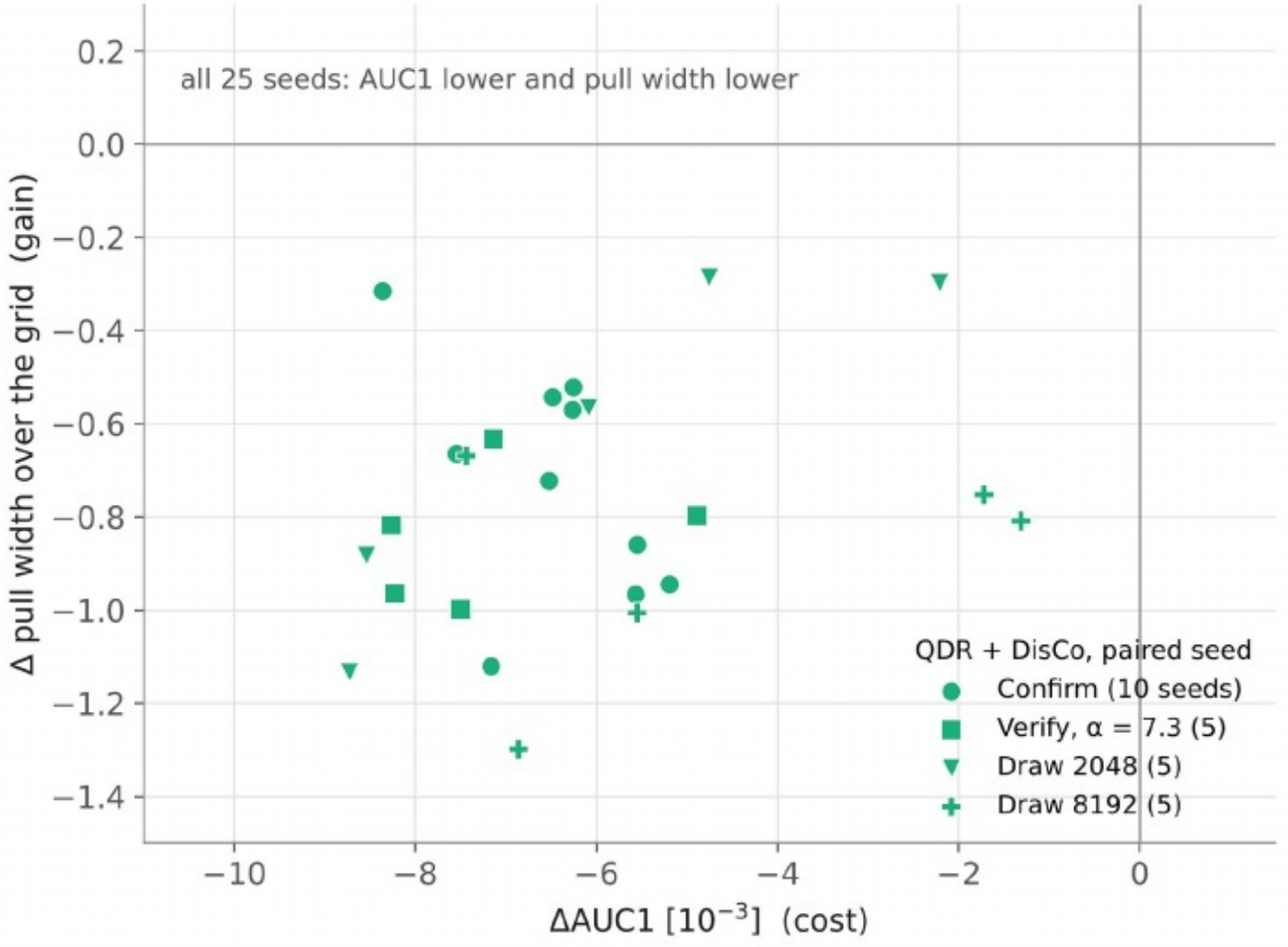
What and where. 7–9% of bkg and 14–17% of signal land in one 0.3-wide NN1 logit window, in all 30 runs, DisCo alone included. In bkg quantile terms it spans 0.79–0.90: on the grid cuts at 0.80 and 0.85, just past the queen's range.

Likely cause (untested). No calo clusters near the vertex puts every cluster input at one value (e.g. MS1Vtx_clusE_raw = 0, a registry input). An isolated vertex looks signal-like, so NN1 may give that population one middling score.

Caveats and next. Only 57% of bkg joined (HTmiss = 0 rows dropped). Zero clusters enriches the spike, it doesn't define it. Next: join through the training loader; fit a depth-3 tree for the exact rule.

Data: R15 confirm, seeds 3000–3009 · plot: DisCo alone, 10 runs · spike counted in all 30 runs (3 arms × 10)
34,970 background events per run · calo-cluster join: 1,711 events in the spike, 7,224 around it (57% of bkg joined)

Every seed pays in AUC and gains in closure



Metric	Pooled change (95% CI)
Pull width over the grid	−0.80 [−0.89, −0.71]
|nc|, queen's range [%]	−1.8 [−2.2, −1.4]
Signed nc (bias) [%]	−2.2 [−2.7, −1.7]
|nc|, working point [%]	−3.0 [−4.5, −1.6]
|nc|, tail q ≥ 0.9 [%]	+3.0 [+1.2, +4.8] worse
AUC1	−0.0065 [−0.0070, −0.0060]

Take-home: the queen buys real closure (in every seed set) for a fixed price of 0.0065 in AUC1, and makes the tail worse. Seeds that pay more don't gain more ($r = 0.24$). Whether the deal is good depends on one test: does raising DisCo's α to the same AUC1 buy as much?

Data: 25 paired seeds (DisCo alone vs QDR + DisCo, same seed) in 4 R15 blocks: confirm 3000–3009 (10), verify 6000–6004 (5), draw 2048 and draw 8192 3100–3104 (5 + 5) → 50 runs · 34,970 bkg + 9,996 sig events per run · 196 cut pairs per plane

Issues, and what would vet or decide each

Issue	What to explore to vet / decide
AUC1 cost of 0.0065; WP signal efficiency −0.011	Decide: a 10-seed DisCo cell at α ≈ 16, matched to AUC1 ≈ 0.871. If DisCo alone buys the same closure there, the queen isn't needed.
NN2 background stretched 3.3× in her range (possible gradient starvation)	Log her soft closure and the hard closure every epoch. If soft improves faster, fix the logit scale she sees (e.g. standardize NN2 logits per batch) and rerun.
Tail +3.0 points worse (pooled); edge +1.2 (confirm tail alone: +0.4, n.s.)	Widen her range to $[0.3, 0.9]^2$ (fits the 4,096 draw); reaching 0.95 needs a draw ≥ 8,192. Tail improves → it was her absence; damage moves out → balloon.
NN1 spike: 7–9% of bkg in one score window at q 0.79–0.90, linked to zero calo clusters; audit covered 57% of bkg	Audit mk2: join through the training loader; depth-3 tree for the exact rule. If it's the encoding, add a has-clusters flag plus a log transform and retrain.
HTmiss exactly 0 for 42.8% of bkg, 0% of signal	Ask Prof. Johns whether 0 is a real value or a fill. Confirm HTmiss is not an NN input and check where it enters the region definitions.

WP = working point: the endcap cut pair, NN1 at bkg quantile 0.70 and NN2 at 0.946 (30% and 5.4% of background pass).

Numbers from R15 endcap: confirm 3000–3009, verify 6000–6004, draw 2048 / 8192 3100–3104 (105 runs graded); HTmiss from the event sample (34,970 bkg, 9,996 sig); NN1-spike audit: confirm, DisCo alone, 10 runs

Not yet shown

1

Does the queen beat simply raising α?

Needs a 10-seed DisCo cell at α ≈ 16, at the same AUC1 (≈ 0.871). This test decides whether the trade is worth making.

2

Is the NN2 stretch an escape?

Log the queen's soft vs hard closure and her per-cut gradient over epochs. If soft closure improves faster than hard closure, the stretch is starving her.

3

Is τ^2 gentler at the edge of her range?

That rests on one 10-seed block. Its AUC cost also depends on draw size: −0.0148 at 2048 vs −0.0011 at 8192.

4

Runs are not bit-reproducible

GPU non-determinism, so every comparison stays within one block against that block's own DisCo reference.

AUC cost of τ^2 by draw size: R15 draw 2048 and draw 8192 blocks, seeds 3100–3104, 5 seeds each (15 runs per block)

UPDATE · 3 NOV 2026

Is the offset real, and what does each queen fix?

The shuffle null, random cuts over the whole plane, and the mk1 queen against mk2.

R15 confirm block: 5 arms × 10 seeds, DisCo $\alpha = 8$, endcap, 34,970 background events per run.

Data: R15 confirm, endcap, seeds 3000–3009 · 5 arms × 10 seeds = 50 runs (50 ABCD planes)
Each run: 34,970 background + 9,996 signal events, out-of-fold

The shuffle null: the offset is real dependence

	Real NN1–NN2 pairs			Shuffle NN2		Shuffle NN1		Shuffle both	
Arm	**mean**	**width**	**±2σ**	**mean**	**width**	**mean**	**width**	**mean**	**width**
DisCo alone	**+2.74**	**2.07**	**41%**	−0.10	0.97	+0.01	1.01	−0.07	0.97
+ queen mk1 (QB)	**+1.71**	**2.11**	**52%**	+0.01	0.95	+0.07	1.01	+0.06	0.96
+ mk2 QDR, $\beta = 0.025$	**+1.80**	**1.50**	**56%**	−0.03	0.98	−0.01	1.00	+0.11	1.00
+ mk2 QDR, $\beta = 0.05$	**+1.61**	**1.32**	**61%**	−0.02	0.97	−0.07	0.99	+0.01	0.96
+ mk2 $\hat{\tau}^2$	**+1.60**	**1.49**	**58%**	−0.06	1.03	+0.04	1.01	−0.06	1.01

Pull $= (N_A - \hat{N}_A)/\sqrt{N_A + \sigma^2_{\hat{N}_A}}$ over the 196 cut pairs of the BASELINE grid, all 10 seeds of each arm. Shuffling one score among background events keeps both score distributions, the counts and the cuts, and only breaks the pairing. 10 shuffles per seed.

Prof. Johns' question: is the offset an artefact of how the cuts are drawn? No. With the pairing broken, the same cuts give a centred N(0, 1) in every arm.

0.0 · 1.0

mean · width of the pull with NN2 shuffled, every arm (statistics: 0, 1)

8–17σ

how far each arm's real mean sits from its null

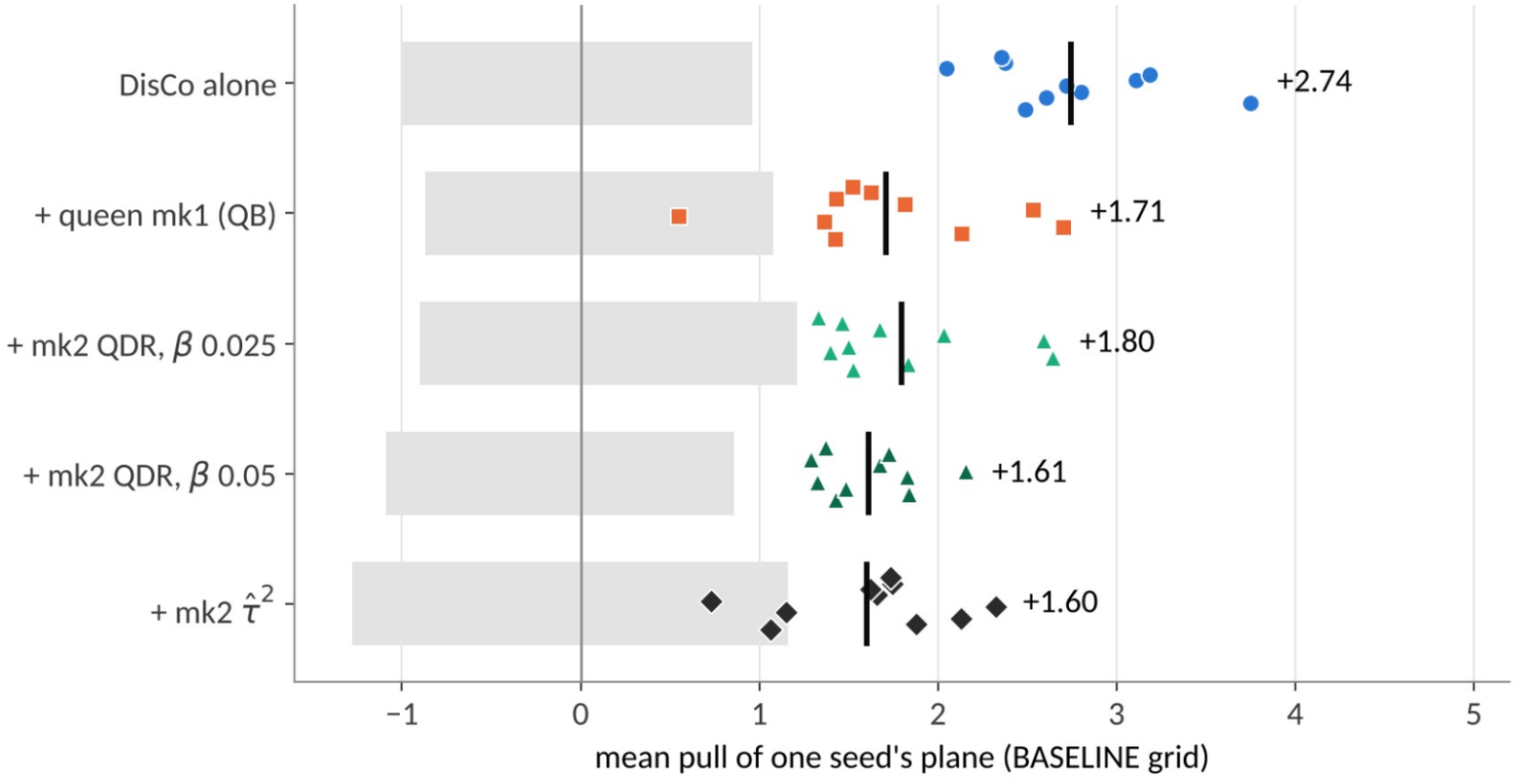
95–97%

shuffled cut pairs within ±2σ (statistics: 95%)

Which score is shuffled does not matter: NN2, NN1 or both give the same null.

Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 seeds = 50 runs · 196 cut pairs per plane · 34,970 bkg events per run
Nulls: 10 shuffles per seed for each of NN2, NN1 and both → 500 shuffled planes per type, 1,500 in all

Every seed's offset sits outside the null



One point per seed, black bar = arm mean. Grey = 95% of shuffled planes for that arm. One plane's cut pairs share events, so its mean pull moves as a single draw (sd ≈ 0.5–0.6 under the null).

47 / 50

seeds outside their arm's null band (DisCo 10, mk1 9, QDR 10 + 10, $\hat{\tau}^2$ 7)

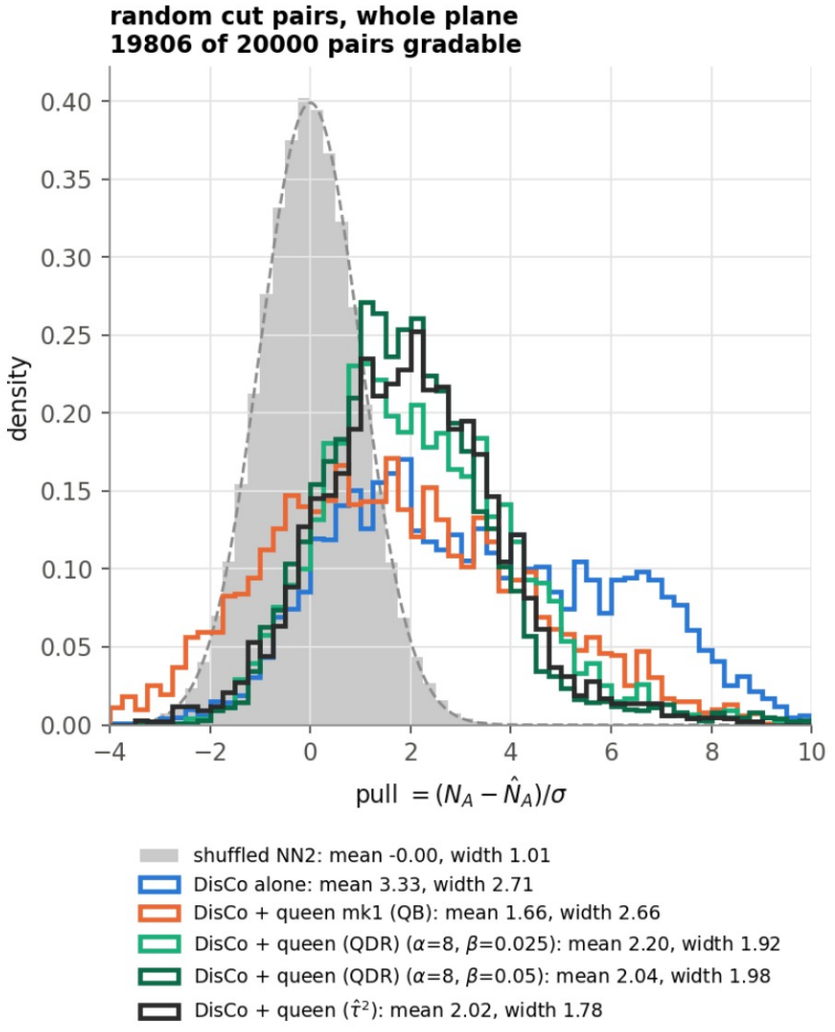
+2.74 → +1.6

mean pull: the queens remove 34–42% of DisCo's offset

The queens move the offset toward zero but no arm reaches it.

Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 seeds = 50 planes · 196 cut pairs per plane · 34,970 bkg events per run
Null band: 500 NN2-shuffled planes, 100 per arm (10 shuffles per seed)

Random cuts over the whole plane: DisCo against both queens



	Random pairs, $q \in [0,1]^2$		Grid $[0.1, 0.9]^2$		Grid $[0.3, 0.8]^2$	
Arm	**mean**	**width**	**mean**	**width**	**mean**	**width**
DisCo alone	+3.33	2.71	+3.41	2.77	+2.95	2.21
+ queen mk1 (QB)	+1.66	2.66	+1.58	2.70	+1.49	2.22
+ mk2 QDR, $\beta = 0.025$	+2.20	1.92	+2.22	1.94	+1.71	1.58
+ mk2 QDR, $\beta = 0.05$	+2.04	1.98	+2.02	1.97	+1.46	1.40
+ mk2 $\hat{\tau}^2$	+2.02	1.78	+2.02	1.83	+1.42	1.53
NN2 shuffled (all arms)	0.00	1.01	+0.01	1.00	0.00	1.01

400 random background-quantile cut pairs per seed (the same pairs for every arm of a seed), 19,806 of 20,000 gradable. $[0.3, 0.8]^2$ is where the mk2 queen draws her cuts.

Fully random cuts do not centre the queens at zero. Outside their training range the offset grows (QDR $\beta = 0.05$: +1.46 inside, +2.04 over the whole plane).

mk1 has the smallest whole-plane mean (+1.66) but DisCo's width (2.66 vs 2.71); mk2 narrows the width to 1.8–2.0.

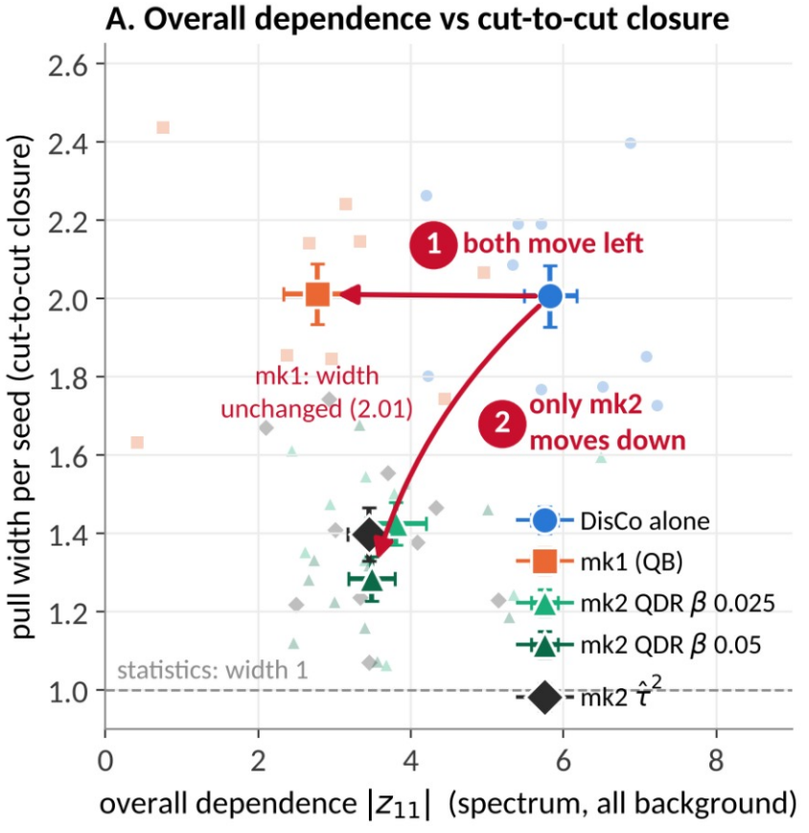
Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 = 50 runs · 400 random cut pairs per run = 20,000 (19,806 gradable)
Grids: 289 points on $[0.1, 0.9]^2$, 121 on $[0.3, 0.8]^2$ per plane · null: 500 NN2-shuffled planes · 34,970 bkg events per run

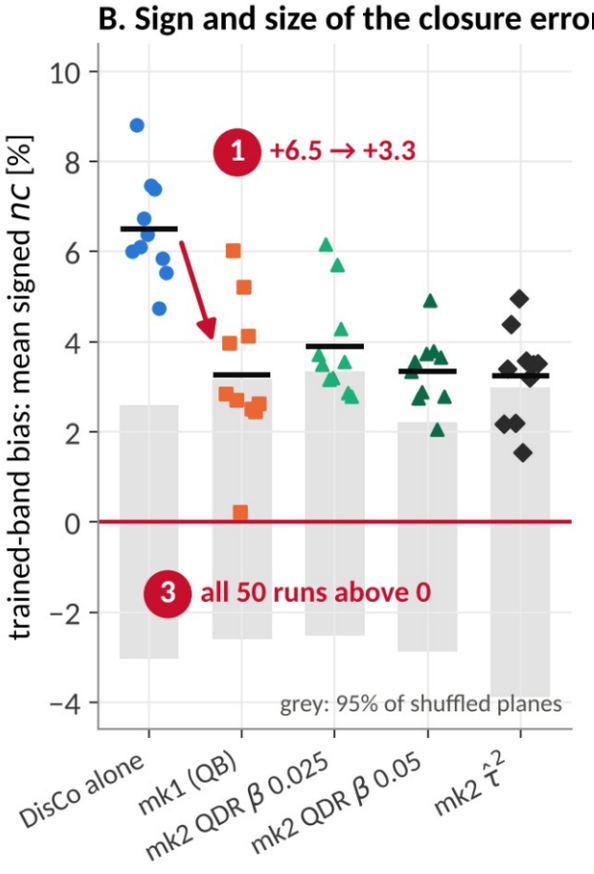
Two queens, two effects

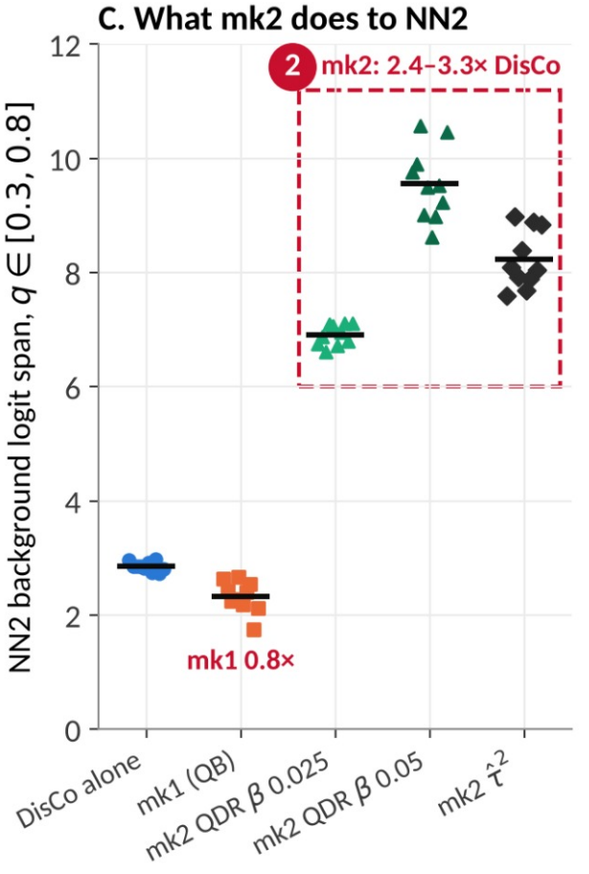
1. **Both queens** halve the overall dependence ($|z_{11}|$ 5.8 → 2.8–3.8) and the bias (+6.5% → +3.3%)

2. **Only mk2** evens out closure across cuts (pull width 2.01 → 1.29–1.43; mk1 stays at 2.01), and only mk2 stretches NN2

3. **Neither** changes the sign: N_A is above the ABCD prediction in all 50 runs







Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 seeds = 50 runs · 34,970 bkg events per run, all used for $|z_{11}|$
Bias null band: 10 NN2 shuffles per seed (100 planes per arm) · pull width over 196 cut pairs per plane

The numbers behind it

Arm	Pull mean	Pull width (pooled)	Pull width (per seed)	Roughness	$\lvert z_{11}\rvert$	Band bias [%]	Band $\lvert nc\rvert$ [%]	NN2 span / DisCo	AUC1
DisCo alone	+2.74	2.07	2.01	0.56	5.83	+6.5	5.95	1.00	0.878
+ queen mk1 (QB)	+1.71	2.11	2.01	0.56	2.77	+3.3	4.15	0.81	0.873
+ mk2 QDR, $\beta = 0.025$	+1.80	1.50	1.43	0.48	3.80	+3.9	3.83	2.42	0.875
+ mk2 QDR, $\beta = 0.05$	+1.61	1.32	1.29	0.46	3.49	+3.3	3.26	3.34	0.871
+ mk2 $\hat{\tau}^2$	+1.60	1.49	1.40	0.49	3.46	+3.3	3.36	2.88	0.872
NN2 shuffled	0.0	1.0	1.0	0.34–0.49	—	±3	1.5–1.7	—	—

Welch t, 10 seeds vs 10	$\lvert z_{11}\rvert$	Pull width per seed	Band bias
mk1 − DisCo alone	−5.4	0.0	−5.1
mk2 QDR ($\beta = 0.05$) − DisCo alone	−5.1	−7.5	−7.2
mk1 − mk2 QDR ($\beta = 0.05$)	−1.3	**+7.6**	−0.1
mk2 $\hat{\tau}^2$ − mk1	+1.3	**−6.0**	0.0

Roughness = mean $\lvert\Delta\,\mathrm{pull}\rvert$ between neighbouring cuts. $\lvert z_{11}\rvert$ = rank-correlation term of the Legendre spectrum on all background, as a z against independence. Band = trained band $[0.3, 0.8]^2$. NN2 span = background logit range over $q \in [0.3, 0.8]$.

The two queens differ on one measure, pull width, by $\lvert t\rvert \approx 6$–8.

Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 seeds = 50 runs · 196 cut pairs per plane · 34,970 bkg + 9,996 sig per run
Welch t: 10 seeds against 10 · nulls: 10 NN2 shuffles per seed

Experimental ABCD-plane health check: the scorecard

Check	DisCo alone	+ queen mk1 (QB)	+ mk2 QDR, $\beta = 0.025$	+ mk2 QDR, $\beta = 0.05$	+ mk2 $\hat{\tau}^2$
Pull mean vs shuffle null	+2.74 · 17σ	+1.71 · 10σ	+1.80 · 10σ	+1.61 · 9σ	+1.60 · 8σ
Pull width (null 1)	2.07	2.11	1.50	1.32	1.49
Band $\lvert nc\rvert$, × null	5.95 · 4.0×	4.15 · 2.7×	3.83 · 2.5×	3.26 · 2.1×	3.36 · 2.1×
Band bias vs null	+6.51%	+3.27%	+3.90%	+3.35%	+3.25%
Seeds with bias > 0	10/10	10/10	10/10	10/10	10/10
Around the WP (3×3), mean pull	+3.25	+2.81	+2.79	+2.33	+2.43
Tail $\lvert nc\rvert$, × null	17.2 · 3.1×	17.9 · 3.6×	15.2 · 2.7×	17.6 · 3.5×	16.7 · 3.4×
Pull-map roughness (null ≤ 0.49)	0.56	0.56	0.48	0.46	0.49
Spectrum $\lvert z_{11}\rvert$	5.83	2.77	3.80	3.49	3.46
NN2 logit span / DisCo	reference	0.81	2.42	3.34	2.88
NN2 shape change (KS / seed-to-seed)	reference	4.9	16.5	19.7	18.3

Experimental check list: abcd_plane_health.py on the 50 confirm runs. Every reference is the same run with NN2 shuffled among background, unless marked. Thresholds on $\lvert nc\rvert$ × null, span and KS are provisional. Data, the same for every arm: HTmiss is exactly 0 for 42.8% of background and 0% of signal; the smallest non-zero background value is 281.

Data: R15 confirm, seeds 3000–3009 · 5 arms × 10 seeds = 50 runs · 196 cut pairs per plane · 34,970 bkg + 9,996 sig per run
Every null: 10 NN2 shuffles per seed (500 planes); per-fold null: same-size shuffle of each fold

Running now: the qtest block

1

Does a signed term remove the bias?

QDR plus $w_b \max(0,\ W\bar{r}^2 - 1)$. $\bar{r}$ pools the cuts' signed residuals before squaring, so a shared offset counts as one effect. w_b = 0.1 and 0.4.

2

Does the tail improve when the queen sees it?

QDR with cuts drawn in $[0.3, 0.95]^2$ instead of $[0.3, 0.8]^2$. Tail $|nc|$ and the spectrum at $q \geq 0.9$ are unchanged by every arm so far.

3

How far do runs drift between blocks?

DisCo alone and QDR re-run on the confirm seeds (3000–3009) as in-block controls. Their gap to confirm measures GPU drift at fixed settings.

4

50 runs, about 1.9 h

5 arms × 10 seeds, $\alpha = 8$, $\beta = 0.05$, 4 workers. Graded with bl_grade, bootstrap and the plane health check. Pre-flight: vectorized = reference on real scores.

Planned: seeds 3000–3009 · 5 arms × 10 seeds = 50 runs · same sample, 34,970 bkg + 9,996 sig events per run